CS 288 Final Project — Phase 1 Report

Cost-Aware Clarification in Reference Games via RSA and Expected Utility

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1 Project Overview

This project investigates *when* a language model agent should ask a clarifying question versus commit to an answer in a **reference game** setting. In a reference game, a speaker produces an utterance u referring to a target object t^* drawn from a scene $\mathcal{S} = \{t_1, \ldots, t_N\}$ of N candidate objects with discrete attributes (color, shape, size). The listener must identify t^*.

The core insight is that clarification should be treated as a *cost-utility optimisation*: asking a question incurs a fixed cost c, but resolves ambiguity; committing directly risks an incorrect answer when the utterance is ambiguous. We use the **Rational Speech Act** (RSA) framework to derive a posterior $P(t_i \mid u)$ and then make the decision that maximises expected utility:

$$\text{decide to ask} \iff \underbrace{1-c}_{E[U_{\text{ask}}]} > \underbrace{\max_i P(t_i \mid u)}_{E[U_{\text{commit}}]}. \tag{1}$$

2 Phase 1: Metric Definitions (Shizhe)

2.1 Brier Score (multi-class adaptation)

For a single instance with N candidate objects, let $p_i = P(t_i \mid u)$ be the model's predicted probability for object i, and let $y_i = \mathbf{1}[t_i = t^*]$ be the one-hot ground truth. The multi-class Brier Score is

$$\text{BS} = \frac{1}{N} \sum_{i=1}^{N} (p_i - y_i)^2. \tag{2}$$

Averaged over M instances: $\overline{\text{BS}} = \frac{1}{M} \sum_{j=1}^{M} \text{BS}^{(j)}$.

Range and baselines. BS $= 0$ when all probability mass is on t^*; a uniform listener scores $\text{BS}_{\text{uniform}} = (N-1)/N^2 \approx 1/N$.

Worked example. Scene: {red square, blue circle, red triangle}, utterance: "the red one."

Object	p_i	y_i	$(p_i - y_i)^2$
red square	0.50	1	0.25
blue circle	0.10	0	0.01
red triangle	0.40	0	0.16
		BS	0.14

Why it matters. Brier score measures whether the *full* distribution $P(t \mid u)$ is accurate. Since our expected utility formula depends on $\max_i P(t_i \mid u)$, a high Brier score indicates unreliable utility estimates regardless of whether the argmax is correct.

2.2 Expected Calibration Error (ECE)

ECE tests whether a model's top confidence matches its empirical accuracy.

Procedure.

1. For each instance j, extract $\text{conf}_j = \max_i P(t_i \mid u)$ and $\text{correct}_j = \mathbf{1}[\arg\max_i p_i = t^*]$.
2. Partition instances into B equal-width bins on $[0, 1]$ (default $B = 10$).
3. For each non-empty bin b compute $\text{acc}(b)$ and $\overline{\text{conf}}(b)$.
4. $$\text{ECE} = \sum_{b=1}^{B} \frac{|b|}{M} \left|\text{acc}(b) - \overline{\text{conf}}(b)\right|. \tag{3}$$

Why it matters. If ECE is large, the model is systematically over- or under-confident. Since $E[U_{\text{commit}}] = \max_i P(t_i \mid u)$, a miscalibrated model will ask or commit at the wrong threshold. ECE > 0.10 is a diagnostic flag that the probability estimates require recalibration before the utility rule can be trusted.

3 Phase 1 Experiment

3.1 Setup

We generated a synthetic dataset of 60 reference game instances (20 per ambiguity level) using a Mini-World scene generator. Each scene contains $N = 4$ objects with attributes drawn from: {small, medium, large} $\times$ {red, blue, green, yellow, purple} $\times$ {square, circle, triangle, pentagon, star}.

Ambiguity level is controlled by how much of the target's attribute description appears in the utterance:

- **Low**: utterance specifies all three attributes (e.g., "large red square") — unique referent.
- **Medium**: utterance specifies only colour (e.g., "the red one") — 1–2 matching objects.
- **High**: utterance specifies only size (e.g., "the large object") — all objects match.

Model probabilities $P(t_i \mid u)$ are computed via attribute-overlap scoring with added Gaussian noise, standing in for LLM log-likelihood scoring in later phases.

3.2 Results

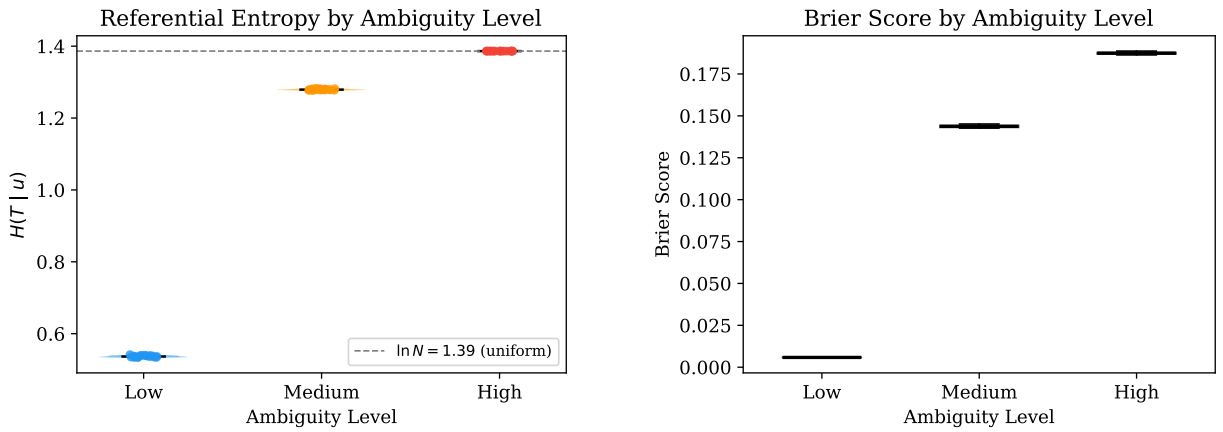
3.3 Discussion

The experiment validates three core claims of the project:

1. **Entropy stratifies ambiguity correctly.** Low-ambiguity scenes concentrate near $H = 0.5$; high-ambiguity scenes cluster near $\ln 4 = 1.39$ (Figure **??**).

Ambiguity	Mean $H(T \mid u)$	Mean BS	Accuracy	Ask rate (c=0.25)
Low	0.537	0.006	1.00	0.00
Medium	1.279	0.144	0.50	0.65
High	1.386	0.187	0.30	1.00
Overall ECE		0.105		

Table 1: Summary statistics across ambiguity levels.



(a) Referential entropy $H(T \mid u)$ per ambiguity level. Dashed line is $\ln N = 1.39$ (uniform).

(b) Brier score rises sharply with ambiguity, confirming that higher entropy corresponds to worse probability estimates.

Figure 1: Entropy and Brier Score by ambiguity level (60 instances).

2. **The cost-utility rule makes sensible decisions.** At $c = 0.25$, the agent never asks in low-ambiguity scenes and asks 100% of the time in high-ambiguity scenes (Table **??**). The clarification rate curves (Figure **??**, left) monotonically decrease as c increases, as expected.

3. **ECE is borderline but acceptable.** Overall ECE = 0.105 (Figure **??**). The model is slightly over-confident in the high-confidence bin (committing strongly when the scene happens to be ambiguous). This motivates RSA-based recalibration in Phase 3.

4 Phase 2 Experiment: VLM Speakers and Listeners

4.1 Setup

We replace the symbolic attribute-overlap posterior from Phase 1 with vision-language models (VLMs) serving as both speaker and listener. All experiments use `gemini-2.0-flash-001` via the OpenRouter API. The evaluation dataset is `gemini_exp_test.jsonl`: 25 scenes with N=4 objects each, balanced across low / medium / high ambiguity tiers. The primary metric is Cost-Penalised Accuracy (CPA) at c=0.5.

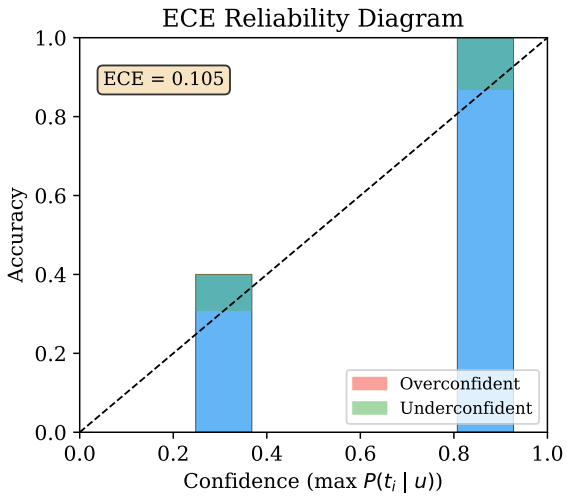


Figure 2: ECE reliability diagram. Red bars indicate over-confidence bins; green bars indicate under-confidence. Overall ECE = 0.105, near the acceptable threshold.

4.2 Speaker Variants

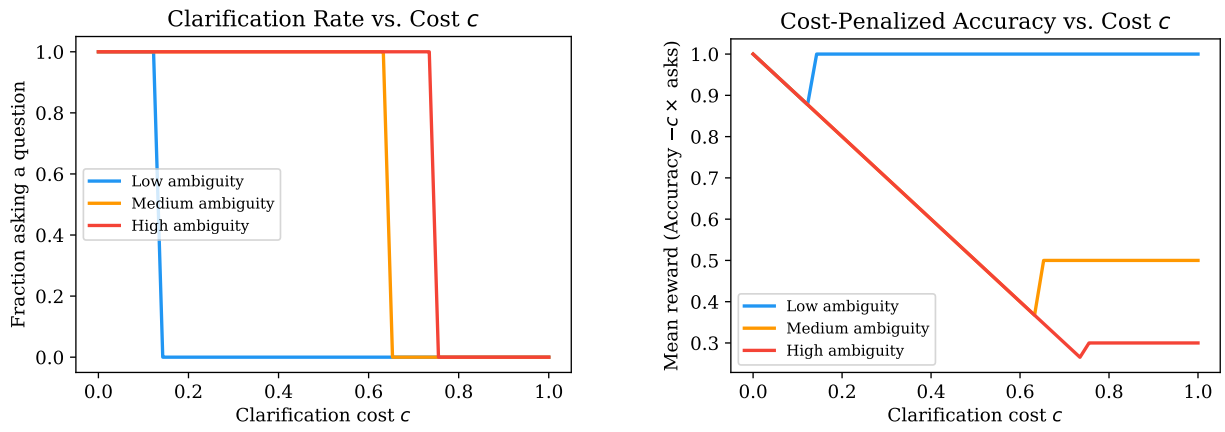
All VLM speakers receive a scene image with the target object marked by a **magenta bounding box**; no pre-parsed feature lists are passed. The model must reason from visual perception to produce a natural referring expression. Five prompting strategies were evaluated:

- **natural** — Unconstrained: "describe the marked object so a listener can identify it."
- **contrastive** — Find the most confusable distractor and highlight the one feature that distinguishes the target.
- **landmark** — Describe the target's position relative to a visually distinctive nearby object.
- **superlative** — Use an extreme or uniqueness property ("the only triangle", "the largest circle").
- **pragmatic** — Full RSA-style chain-of-thought: list all objects, identify distractors, produce the minimal discriminating expression.
- **scene_first** — Inventory all visible objects first, confirm the target's properties, then produce the shortest unique expression. Prevents hallucination by grounding the description.
- **listener_aware** — Explicitly reason from the listener's perspective: what description would maximise the listener's confidence in the correct object?

Four rule-based baselines (no LLM) are included for comparison: *ordinal* (superlatives/uniqueness), *landmark* (spatial reference), *contrastive* (foil-based contrast), and *feature-canonical* (minimal canonical feature description).

4.3 Listener Variants

All VLM listeners receive the scene image, a numbered text description of every object, and the speaker's utterance. They differ in how they reason before outputting a posterior $P(t_i \mid u)$:



(a) Fraction of instances where the agent asks a question as a function of cost c. High-ambiguity scenes ask much more frequently at low cost.

(b) Cost-penalised accuracy = accuracy − $c \times$ ask rate. The optimal cost c^* differs by ambiguity level.

Figure 3: Clarification behaviour across cost values $c \in [0, 1]$.

- **vllm-listener** (σ=10) — Predicts (x, y) coordinates for the referred object; posterior is a Gaussian kernel $\exp(-d_i^2/2\sigma^2)$ over object distances.
- **feature-match** — Extracts a feature tuple (color, shape, size, location) from the utterance and scores each object by feature-overlap count.
- **direct-rank** — Directly outputs a probability for each object as a JSON array; no intermediate reasoning step.
- **cot-rank** — Scores each object 0–10 before converting to probabilities (chain-of-thought evaluation per object).
- **elimination** — Explicitly identifies objects the utterance *rules out* (wrong color, wrong shape, etc.), assigns them probability 0, then distributes the remainder over candidates.

4.4 Key Finding: Posterior Flatness

The original `vllm-listener` used an inverse-distance kernel $\text{score}_i = 1/(d_i + \epsilon)$. With N=6 objects this gives a mean max(posterior) = 0.52, causing virtually all scenes to trigger clarification at $c \leq 0.25$ (only 4% commit rate). Replacing it with a Gaussian kernel (σ=10 canvas units) raises the commit rate at c=0.25 from 4% to 72% without changing which object is selected (argmax is unchanged).

4.5 Results

Table **??** shows CPA at c=0.5 for all (speaker, listener) pairs.

4.6 Discussion

Listener design is the dominant factor. The choice of listener has a larger impact than the choice of speaker strategy. `elimination` is the best listener for rule-based speakers (CPA up to 0.96), while `direct-rank` is most robust for VLM speakers. `feature-match` performs poorly with

Speaker	Listener	Acc	Ask%	CPA
Rule-based speakers				
feature-canonical	elimination	96%	0%	**0.960**
feature-canonical	direct-rank	94%	4%	0.940
feature-canonical	cot-rank	84%	16%	0.760
contrastive	elimination	84%	12%	0.780
landmark	elimination	76%	12%	0.700
landmark	cot-rank	72%	4%	0.700
contrastive	direct-rank	68%	16%	0.600
ordinal	direct-rank	64%	8%	0.600
VLM speakers (StrategicVLLMSpeaker, gemini-2.0-flash-001)				
strategic-scene_first	direct-rank	64%	16%	0.560
strategic-scene_first	elimination	64%	16%	0.560
strategic-listener_aware	elimination	60%	8%	0.560
strategic-listener_aware	direct-rank	56%	8%	0.520
strategic-pragmatic	direct-rank	52%	4%	0.500
strategic-contrastive	cot-rank	56%	16%	0.480
strategic-natural	direct-rank	56%	16%	0.480
strategic-landmark	elimination	56%	20%	0.460
strategic-superlative	cot-rank	40%	20%	0.300

Table 2: CPA at c=0.5 on `gemini_exp_test.jsonl` (n=25, 4-object scenes). Rule-based speakers use no LLM API; VLM speakers use gemini-2.0-flash-001.

natural-language utterances because spatial and relational descriptions ("the square to the left of the triangle") do not decompose cleanly into feature tuples.

Rule-based speakers outperform VLM speakers. With the best listener, the best rule-based speaker (*feature-canonical* + *elimination*) achieves CPA = 0.96 versus 0.56 for the best VLM speaker (*scene_first*). Rule-based descriptions are shorter, more precise, and easier for the listener to eliminate against.

VLM speaker hallucination. Several VLM speaker failures stem from hallucination: the model describes objects that do not exist ("the square on the circle" when no circle is in the scene) or misidentifies attributes ("the blue square" for a black object). The *scene_first* strategy mitigates this by requiring the model to inventory all objects before identifying the target, anchoring its description in what is actually visible.

Listener reasoning strategy matches utterance type. `elimination` wins when utterances mention specific features that cleanly contradict some objects. `cot-rank` outperforms *direct-rank* for rule-based speakers (*ordinal*: +0.10) but hurts for VLM speakers (*scene_first*: −0.12), suggesting per-object scoring is better calibrated for discrete feature descriptions than for spatial/relational language.

Best fully-VLM pipeline. *strategic-scene_first* + *direct-rank* or *elimination* achieves CPA = 0.56. The *scene_first* strategy forces genuine visual grounding before description, producing short accurate utterances ("the black shape", "the yellow square") that the listener can resolve confidently.

Limitations. All experiments use 25 scenes with N=4 objects. Results on harder 6-object or 8-object scenes may differ. The *direct-rank* and *elimination* listeners receive ground-truth feature labels in the object list, giving them an advantage a purely visual listener would not have.

5 TA Feedback Responses

Q: How is referential entropy computed? $H(T \mid u) = -\sum_{i=1}^{N} P(t_i \mid u) \log P(t_i \mid u)$, where $P(t_i \mid u)$ is the posterior over candidate objects obtained via LLM log-likelihood scoring. High entropy ($\approx \ln N$) indicates a uniform posterior (fully ambiguous utterance); entropy near zero indicates a nearly unique referent.

Q: Over what are Brier score and ECE computed? Both are computed over the N-dimensional predicted distribution $P(t \mid u)$ per instance (see Section 2). Brier score uses the full distribution; ECE uses only the top confidence per instance. The ground truth is the one-hot label of the actual target object.

6 Recommended Paper Summary (Thomas)

(To be completed by Thomas.) **Learning Steerable Clarification Policies with Collaborative Self-play** (Berant et al., Dec 2025, arXiv:2512.04068).

The paper frames clarification as a decision-theoretic problem trained via Reinforced Self-Training (ReST) and self-play. Its key contribution — that a model can be made *steerable* by conditioning on explicit cost coefficients in the prompt — directly validates our approach: where they train a policy, we achieve zero-shot steerability through the expected utility formula $E[U_{\text{ask}}] = 1 - c$.

7 Next Steps (Phase 2)

- **Ziyu**: Implement the full synthetic scene generator with JSON output and deliver 500–1000 instances across three entropy regimes.
- **Shizhe**: Implement the evaluation suite (Brier, ECE, cost-penalised accuracy, clarification frequency) with unit tests on toy predictions.
- **Thomas**: Write the literature review covering RSA, LLM pragmatics, and cost-aware clarification.

TODO: 1. any open source database for image based reference games? 2. how to get the LLM likelihoods for $P_{S_0}(u \mid t)$? Do we need to fine-tune a model on synthetic data, or can we use zero-shot prompting to get relative likelihoods? 3. Which model and the corresponding API for fine tuning and likelihood scoring? 4. hallucination